Syllabus: Creating your own (quantitative) research project in R

Version: 1

You can find the latest version of the syllabus on the project here.

Introduction

This course wants to enable students to run their own (quantitative) research project in R. In the process, they will learn basic data wrangling and data visualisation in R. Every student will choose a project that they want to conduct using publicly available data of their choosing (some examples will be provided) and will, in consultation with the course coordinator, work towards a project report at the end of the course that they will present to the other students. Students will analyse their data in the open-source programme R and will be encouraged to write their final reports in it, too. The course will be a mixture of lectures on practical issues and a project based working sessions in exchange with the course coordinator and other students to make use of peer-to-peer learning. The lecture will be tailored to the needs and levels of the participating students.

In this course students are encouraged to run their own small-scale analysis project using R from beginning to end: choosing a topic and research question they find interesting, selecting and downloading the relevant data, using simple (presumably mostly) descriptive statistics such as means, and writing a report about their findings. They will be encouraged to use Quarto/ Rmarkdown, the integrated tool within R-Studio, to write their final reports and export them into PDF to automatically integrate updated findings. Also, students will be encouraged to upload and maintain their project on git. This course teaches practical skills and combines it with theoretical grounding in reproducible science. Hence it can be used as a good foundation for quantitative bachelor or master theses, and to learn skills that can be used also outside of academia.

Assessment

Students can take the course for 3 CP or 6 CP. The final assessment will depend on the amount of CP students chose. **Both assessments** include the **simple progress report every two weeks** (just about one paragraph in an unstructured form, including reasons for no progress, which will always happen, is no problem and part of the process) and the **final script** of the project. Beyond that, the two different levels encompass the following two assessments:

3 Credit Points (CP):

- Short presentation (10 minutes)
- Short final research note with references (~5 pages, at least one high-quality graph)

6 Credit Points (CP):

- Long presentation (20 minutes)
- Long final research note with references (~10 pages, at least two high-quality graphs)

Prerequisites

- Basic understanding of statistics (means, confidence intervals), for instance through Statistics 1 at Bremen University
- Preferably but not necessarily some basic experience in R or alternative statistics software (Stata, Python, SAS etc.)
- Laptop that students can bring to class, preferably pre-installed with R, R-studio and Quarto

Time and Location

Time: Monday 14:00 - 16:00

Location: Socium, Mary-Somerville-Str. 3, Room 3.3390

Course schedule (preliminary)

This is the preliminary course schedule. It may change based on student demand.

~~April 6th:~~ Easter

April 13th: Introduction and installing R and Rstudio, Quarto and Github

April 20th: Choosing a research project

April 27th: (Short) introduction to Rstudio and Quarto

May 4th: Reproducible science (and git)

May 11th: The tidyverse: What is it and how can you use it?

May 18th: Good practices when coding

~~May 25th:~~ Whit Monday

June 1st: Crash course in data wrangling

June 8th: Introduction to data visualisation in ggplot

June 15th: More data visualisation in ggplot

June 22nd: Presentations

June 29th: Presentations

July 6th: Final class

Notes:

The ~~struck through~~ lessons are missing due to holidays.

The underlined lessons are weeks with progress reports due.

Instructor

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Office: 5.4010 (floor above the seminar room)

Office days: Monday and Wednesday

Inspiration:

- Gentzkow, M. and Shapiro, JM. (2014): Code and Data for the Social Sciences: A Practitioner's Guide https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf
- Munafò, Marcus R., Brian A. Nosek, Dorothy V. M. Bishop, et al. 'A Manifesto for Reproducible Science'. Nature Human Behaviour 1, no. 1 (2017): 0021. https://doi.org/10.1038/s41562-016-0021.
- Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 'Introduction'. In R for Data Science, 2nd Edition, Second edition. O'Reilley, https://r4ds.hadley.nz/intro.html.